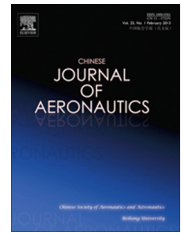




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A new remaining useful life estimation method for equipment subjected to intervention of imperfect maintenance activities

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Abstract As the key part of Prognostics and Health Management (PHM), Remaining Useful Life (RUL) estimation has been extensively investigated in recent years. Current RUL estimation studies considering the intervention of imperfect maintenance activities usually assumed that maintenance activities have a single influence on the degradation level or degradation rate, but not on both. Aimed at this problem, this paper proposes a new degradation modeling and RUL estimation method taking the influence of imperfect maintenance activities on both the degradation level and the degradation rate into account. Toward this end, a stochastic degradation model considering imperfect maintenance activities is firstly constructed based on the diffusion process. Then, the Probability Density Function (PDF) of the RUL is derived by the convolution operator under the concept of First Hitting Time (FHT). To implement the proposed RUL estimation method, the Maximum Likelihood Estimation (MLE) is utilized to estimate the degradation related parameters based on the Condition Monitoring (CM) data, while the Bayesian method is utilized to estimate the maintenance related parameters based on the maintenance data. Finally, a numerical example and a practical case study are provided to demonstrate the superiority of the proposed method. The experimental results show that the proposed method could greatly improve the RUL estimation accuracy for the degrading equipment subjected to imperfect maintenance activities.

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1. Introduction

With the increasing requirement for the safety and reliability of engineering equipment, Prognostics and Health Management (PHM) technique has received widespread attention in the academic and industrial field during the past decades.^{1–3} As the key part of PHM, Remaining Useful Life (RUL) estimation can be used to determine the maintenance time,

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Nomenclature

n	upper limit of the preventive maintenance (PM) number	$B(\bullet)$	Brownian motion
ω_p	PM threshold	i	PM number before $t_{i,j}$
ω	failure threshold	j	CM number after the i th maintenance
T_i	time of the i th maintenance, $i \leq n$, $i \in \mathbf{N}^+$	$t_{i,j}$	the j th monitoring time after the i th maintenance
r_i	condition monitoring (CM) number between T_i and T_{i+1}	$L_{i,j}$	RUL at time $t_{i,j}$ with its realization $l_{i,j}$ and PDF $f_{L_{i,j}}(l_{i,j})$
η_i	residual degradation coefficient	$L_{n,j}$	RUL at time $t_{n,j}$ with its realization $l_{n,j}$ and PDF $f_{L_{n,j}}(l_{n,j})$
a, b	hyperparameters in η_i	$R_{i,j}$	remaining operating time of the $(i+1)$ th stage with its realization $r_{i,j}$ and PDF $f_{R_{i,j}}(r_{i,j})$
t	time	R_{i+k}	operating time of the $(i+k+1)$ th stage with its realization r_{i+k} and PDF $f_{R_{i+k}}(r_{i+k})$
$\int_0^{t-T_i} \mu(\tau, \theta) d\tau$	nonlinear degradation function with its parameter θ and integration variable τ	R_n	operating time of the $(n+1)$ th stage with its realization r_n and PDF $f_{R_n}(r_n)$
c_i	degradation rate changing factor after the i th maintenance	m	total number of the historical equipment
μ_c, σ_c	parameters in the distribution of c_i	ℓ	likelihood function
σ_B	diffusion coefficient		

inspection interval, or spare parts' ordering quantity by minimizing economic cost or mitigating failure risk,²⁻⁷ and hence, increasing attention has been paid to the RUL estimation domain. The remaining useful life of an asset is generally defined as the length from the current time to the end of the useful life.² The existing RUL estimation studies can be grouped into the following two types: physics-based methods and data-driven methods.¹ For these two types of methods, physics-based methods are based on the identification of potential failure mechanisms for a device, product, or system, and thus the failure can be easily located and insulated. However, due to the complexity of the engineering system as well as the diversity and uncertainty of the operating environment, it is extremely difficult to identify the potential failure mechanisms. Owing to the rapid development of sensory technology, data-driven methods have become the mainstream in the current RUL estimation domain.

The commonly-used data-driven methods include Wiener process, Gamma process, and Markov chain.² Because Wiener process can be adopted to describe both the monotonous and non-monotonous degradation processes, it exhibits better performance compared with Gamma process and Markov chain in many engineering practice including rotating element bearings, self-regulating heating cables, laser generator and bridge beams. Therefore, Wiener process based degradation models have been studied intensively in literature. Based on Wiener process with a linear drift, Tseng et al. determined the lifetime of the light intensity of LED lamps in Ref.⁸. Ye et al. investigated linear-drifted Wiener process with measurement error and developed a mixed effects model motivated by the degradation process of magnetic heads used in hard disk drives.⁹ Due to the extensive existence of the nonlinear degradation patterns, a nonlinear-drifted Wiener process model as well as an analytical RUL distribution derivation method was presented in Ref.¹⁰. Wang et al. further proposed an additive Wiener process based RUL estimation method for the hybrid deteriorating systems, where the stochastic dependencies between different deteriorating patterns were considered.¹¹ Based on Wiener process with a nonlinear drift, Huang et al. presented a recursive filter algorithm, which could improve

the estimation accuracy of the RUL.¹² Recently, a nonlinear-drifted Wiener process modeling method was proposed for the RUL estimation of rechargeable batteries by taking multiple hidden states into consideration.¹³

Most of the existing RUL studies assumed that the degrading equipment was not maintained during its life cycle.⁴⁻¹⁴ Nevertheless, in practice, when the degradation level of the equipment exceeds a specified threshold, maintenance activities are often carried out for the equipment to improve the performance. According to the effect of maintenance, maintenance activities can be divided into perfect maintenance, imperfect maintenance, and minimal maintenance.¹⁵ Specifically, perfect maintenance can restore the equipment to an as-good-as new state, and minimal maintenance can restore the equipment to an as-bad-as old state. Differently, imperfect maintenance can restore the equipment to the state between as good as new and as bad as old.¹⁶⁻²³ Therefore, compared with perfect maintenance and minimal maintenance, imperfect maintenance is more general in the engineering practice, examples of imperfect maintenance are spraying lubricant for drill bit, adjusting the dynamic balance of the fan.²¹⁻²³ Imperfect maintenance activities can normally slow down the degradation of the equipment, thereby prolonging the life of the equipment.¹⁵ If the influence of the imperfect maintenance is not well taken into account, the RUL estimation for the equipment subjected to the intervention of imperfect maintenance activities may have large biases. Thus, imperfect maintenance activities have a critical influence on the RUL estimation of such degrading equipment, and it is important to account for the influence of the imperfect maintenance activities when we estimate the RUL of the equipment in such situations.²¹⁻²³

However, although imperfect maintenance has attracted great attention in maintenance activities scheduling area,²⁴⁻²⁶ few studies could be found in literature to consider the influence of imperfect maintenance activities on RUL estimation. You and Meng proposed an extended proportional hazards model to carry on the simulation analysis for the maintenance activities,²⁷ which could obtain the mean and variance of the RUL, but could not obtain the distribution of the RUL. Wang et al. used Wiener process with negative jumps to estimate the RUL

of the equipment subjected to imperfect maintenance activities,²² which only considered the influence of maintenance activities on the degradation level but ignored the influence of maintenance activities on the degradation rate. Contrarily, Zhang et al. estimated the RUL of the equipment under the consideration of influence of maintenance activities on the degradation rate, but ignored the influence of maintenance activities on the degradation level.²³ In the engineering practice, it is well recognized that imperfect maintenance activities have a certain influence on both the degradation level and the degradation rate.^{21,23,28} For example, as a commonly-used imperfect maintenance strategy, welding process for metal component can reduce the crack length, but it can also destroy the physical mechanism of inner materials and therefore accelerate the degradation rate of the metal component. It should be noted that, although new products also have heterogeneous phenomenon and different degradation rates,^{9–11} this is different with what we will consider in this manuscript in that the degradation rate is assumed to become larger after maintenance. In order to appropriately characterize the influence of the imperfect maintenance activities and accurately estimate the RUL of the equipment, these two factors should be taken into account at the same time.

Aimed at the aforementioned problem in current RUL estimation studies of the degrading equipment subjected to imperfect maintenance activities, this paper proposes a new degradation modeling and RUL estimation method considering the influence of imperfect maintenance activities on both the degradation level and the degradation rate. Compared with the similar studies in literature (such as Wang et al.²² and Zhang et al.²³), the main contributions of this paper are twofold: (1) considering the influence of imperfect maintenance activities on both the degradation level and the degradation rate; (2) deriving the Probability Density Function (PDF) of the RUL for the equipment subjected to imperfect maintenance by the convolution operator. Specifically, a stochastic degradation model considering the influence of imperfect maintenance activities is firstly constructed based on the diffusion process. Based on the constructed model, the PDF of the RUL is derived by the convolution operator under the concept of First Hitting Time (FHT). To implement the presented method, the unknown model parameters are estimated by the maximum likelihood estimation method and Bayesian method based on the available Condition Monitoring (CM) data and maintenance data. The results of a simulated example and a practical case are provided to demonstrate the superiority of the proposed method.

The remainder of the paper is organized as follows. Section 2 gives the problem description and associated assumptions. Section 3 presents the degradation modeling method based on the diffusion process and derives the PDF of the RUL. The parameter estimation process is given in Section 4. A numerical example is provided in Section 5 for demonstration. Section 6 provides a practical case study for illustration. The main conclusions are summarized in Section 7.

2. Problem description and assumptions

2.1. Problem description

For the engineering equipment, the degradation level can be obtained by CM technique, and it is compared with the

Preventive Maintenance (PM) threshold to make the maintenance decisions. When the degradation level exceeds the preset PM threshold, the maintenance activities will be carried out. Considering the effectiveness of PM, we can further classify the PM as perfect maintenance, minimal maintenance and imperfect maintenance. As mentioned above, due to the generalization of imperfect maintenance activity relatively to the other two PM strategies, we will focus on this kind of maintenance scheme in this paper. Specifically, the imperfect maintenance activities can not only reduce the degradation level of the system, but also increase the system's degradation rate. In order to ensure the operating reliability and lower the maintenance cost, the upper limit of the total number of maintenance activities is assumed to be finite, denoted as n . In other words, when the number of the imperfect maintenance reaches the specified n , the engineering equipment will no longer be maintained, and it will be totally replaced before its degradation level first hits the specified failure threshold.

Fig. 1 provides a schematic diagram of the degradation trajectory considered in this paper, which characterizes the continuous degradation process of the engineering equipment subjected to the intervention of imperfect maintenance activities. In Fig. 1, the PM threshold of the equipment is denoted by ω_p and the failure threshold is denoted by ω . $T_i (i \leq n, i \in \mathbb{N}^+)$ denotes the time of the i th maintenance, and T_{n+1} denotes the failure time. We use $t_{i,j}$ to represent the CM time, where i is the maintenance number before $t_{i,j}$, $j (j \in [0, r_i])$ is the CM number after the i th maintenance, and r_i is the total CM number between T_i and T_{i+1} . It can be readily found that the time instant denoted by $t_{i-1,r_{i-1}}$, T_i , and $t_{i,0}$ is equal to each other, i.e., $t_{i-1,r_{i-1}} = T_i = t_{i,0}$.

As shown in Fig. 1, the degradation process can be divided into $n+1$ stages during the whole life cycle by the total number n of imperfect maintenance activities, i.e., $(0, T_1), (T_1, T_2), \dots, (T_n, T_{n+1})$. We assume that every stage of the degradation process is independent with each other. It can be intuitively seen that the life T of the equipment subjected to the intervention of imperfect maintenance activities can be regarded as the sum of the operating time of each stage, i.e., $T = \sum_{i=0}^n R_n$, where R_i represents the operating time of the $(i+1)$ th stage.

Based on the above description, we will study the following problems in this paper:

- (1) How to construct the degradation model of the equipment subjected to the intervention of imperfect maintenance activities;
- (2) How to derive the PDF of the RUL at any time for the equipment subjected to the intervention of imperfect maintenance activities;
- (3) How to estimate the parameters in the degradation model based on the CM data and the maintenance data.

2.2. Assumptions

On the basis of the problem described above, we propose the assumptions required either for practical consideration, or for modeling simplification.

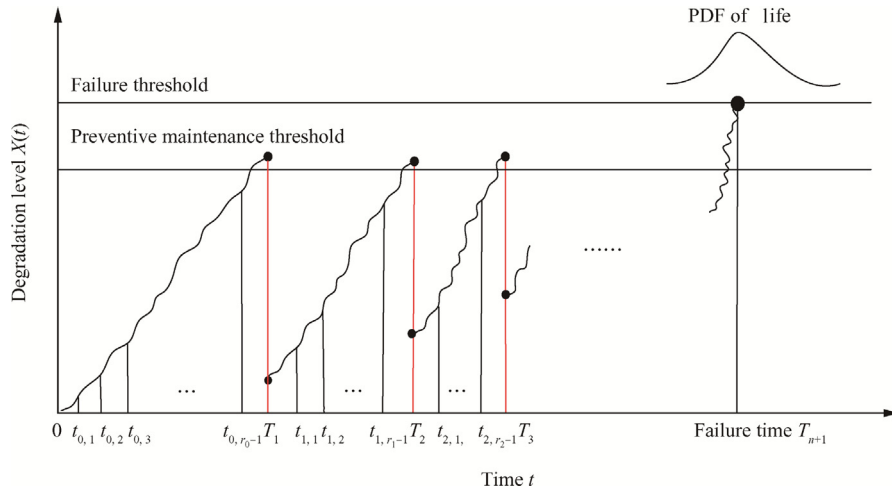


Fig. 1 Degradation trajectory under intervention of maintenance activities.

- (1) The equipment is monitored perfectly and periodically.
- (2) The PM is imperfect and can restore the health condition of the equipment to a state between as good as new and as bad as old.
- (3) The total number of the PM is finite. Specifically, when the maintenance number for the equipment reaches the specified upper limit n , the PM activities will not be carried out any longer due to the reliability and cost considerations.
- (4) If the degradation level is between the PM threshold ω_p and the failure threshold ω , a PM is carried out immediately.
- (5) If the degradation level exceeds the failure threshold ω , the equipment is required to be replaced immediately.
- (6) If the degradation level is less than the PM threshold, the equipment is left unchanged until the next CM time.

Assumption 1 is common in degradation modeling practice.^{2,3,11} Assumption 2 is due to the consideration for the engineering practice and is widely accepted in the academic field.^{15–17} It is noted that assumptions 3–6 have been widely adopted in maintenance and inventory modeling practices for modeling simplification (see for examples in Refs. ^{21,23,28}).

3. Degradation modeling and RUL estimation

3.1. Degradation modeling

In this paper, diffusion process is utilized to describe the nonlinear stochastic degradation under the intervention of imperfect maintenance activities, i.e., the degradation path of the equipment can be modeled by diffusion process at each stage between two successive maintenances. The degradation level after the i th maintenance activity can be denoted as

$$X(t) = \eta_i \omega_p + \lambda_i \int_0^{t-T_i} \mu(\tau, \theta) d\tau + \sigma_B B(t - T_i) \quad (1)$$

where $X(t)$ is the degradation level at time t ; $i(i \leq n, i \in \mathbb{N}^+)$ is the maintenance number performed before t ; η_i is the residual degradation coefficient after the i th maintenance activity, which is assumed to follow a normal distribution, i.e.,

$\eta_i \sim N(1 - \exp(-ai), b)$, and a and b are hyperparameters²⁹; ω_p represents the PM threshold of the equipment specified by industrial standard and expert experience; $\mu(\tau, \theta)$ is used to characterize the nonlinear stochastic nature, τ is the integral variable, and θ is the fixed parameter related to the degradation rate; λ_i is a random parameter related to the degradation rate. From the mathematical point, the degradation rate can be regarded as the derivative of the degradation level. To describe the influence of maintenance activities on the degradation rate, we introduce the degradation rate changing factor c_i as the maintenance parameter and assume that λ_i is formulated as $\lambda_i = c_i \lambda_0$, where λ_0 denotes the fixed parameter related to the degradation rate, and c_i is the degradation rate changing factor after the i th maintenance activity. The PDF of c_i is denoted by $f(c_i|i, \mathbf{Y})$, where $\mathbf{Y}$ is the parameter vector in the distribution of c_i . In particular, $c_0 = 1$. T_i is the time of the i th maintenance, σ_B is the diffusion coefficient, and $B(\cdot)$ is the standard Brownian Motion (BM).

Remark 1. The reasons that the residual degradation coefficient is assumed to follow the specified normal distribution are twofold. First, it is easier to obtain the analytical results or it can facilitate the derivation process under the normal distribution, which is desirable from both the theoretical and practical angles. Second, it is noted that the normal distribution has plenty of good characteristics, whose mean and variance for the residual degradation coefficient had been given to describe imperfect maintenance model in Ref. ²⁹.

Remark 2. The reason that the degradation rate changing factor c_i after the i th maintenance activity is used to describe the influence of imperfect maintenance activities on the degradation rate is mainly to facilitate derivation, which is modeled by Zhang et al.²³

3.2. Derivation of PDF of RUL

Now, we discuss how to estimate the RUL for the degrading equipment defined in Eq. (1). Based on the concept of FHT, the RUL at time t_{ij} is defined as^{10,11}

$$L_{i,j} = \inf\{l_{i,j} : X(t_{i,j} + l_{i,j}) > \omega | x_{i,j} < \omega\} \quad (2)$$

where $L_{i,j}$ is the RUL at time $t_{i,j}$ with its realization $l_{i,j}$, and $x_{i,j}$ is the degradation level at time $t_{i,j}$.

Based on the analysis in Section 2, the degradation process can be divided into $n + 1$ stages. In order to illustrate the RUL intuitively and derive the PDF of the RUL theoretically, we need to define the following variables: $L_{n,j} = \inf\{l_{n,j} : X(t_{n,j} + l_{n,j}) > \omega | x_{n,j} < \omega\}$ denotes the RUL at time $t_{n,j}$, $R_{i,j} = \inf\{r_{i,j} : X(t_{i,j} + r_{i,j}) > \omega_p | x_{i,j} < \omega_p\}$ denotes the remaining operating time at time $t_{i,j}$ for the $(i + 1)$ th stage, $R_{i+k} = \inf\{r_{i+k} : X(t_{i+k,0} + r_{i+k}) > \omega_p | x_{i+k,0} < \omega_p, 1 \leq k < n - i\}$ denotes the operating time of the $(i + k + 1)$ th stage, and $R_n = \inf\{r_n : X(t_{n,0} + r_n) > \omega | x_{n,0} < \omega\}$ denotes the operating time of the $(n + 1)$ th stage. Thus, as shown in Fig. 2, the RUL $L_{i,j}$ at time $t_{i,j}$ can be represented as

$$L_{i,j} = \begin{cases} L_{n,j} & i = n \\ R_{i,j} + \sum_{k=1}^{n-i} R_{i+k} & 0 \leq i < n \end{cases} \quad (3)$$

According to Eq. (3) and Fig. 2, the RUL estimation for the equipment subjected to the intervention of imperfect maintenance activities can be divided into two cases, i.e., $i = n$ and $0 \leq i < n$. We will derive the PDF of the RUL for the equipment in the following section.

Based on the work by Gnedenko and Kolmogorov,³⁰ the PDF of the RUL can be further denoted by

$$f_{L_{i,j}}(l_{i,j}) = \begin{cases} f_{L_{n,j}}(l_{n,j}) & i = n \\ f_{R_{i,j}}(r_{i,j}) \otimes f_{R_{i+1}}(r_{i+1}) \otimes \cdots \otimes f_{R_n}(r_n) & 0 \leq i < n \end{cases} \quad (4)$$

where $f_{L_{i,j}}(l_{i,j})$ is the RUL at time $t_{i,j}$, $f_{L_{n,j}}(l_{n,j})$ is the RUL at time $t_{n,j}$, $f_{R_{i,j}}(r_{i,j})$ is the PDF of the remaining operating time for the $(i + 1)$ th stage, $f_{R_{i+k}}(r_{i+k})$ is the PDF of the operating time for the $(i + k + 1)$ th stage, $f_{R_n}(r_n)$ is the PDF of the operating time for the $(n + 1)$ th stage, and $\otimes$ is the symbol of convolution operator.

In order to derive the PDF of the RUL, it is necessary to obtain the PDFs of the defined variables above. We assume that the degradation rate changing factor c_i follows a normal distribution for simplification, i.e., $c_i \sim N(i\mu_c, \sigma_c^2)$, where the parameters in the distribution is denoted by $\mathbf{Y} = (\mu_c, \sigma_c^2)$, and obviously, the mean of stochastic parameter c_i is growing with the increase of the maintenance number i .

Based on the above analysis, the PDF of the RUL can be theoretically derived for these two cases.

Case 1. When $i = n$, according to the RUL described in Eq. (3), we can derive the conditional PDF of the RUL at time $t_{n,j}$ on condition that the stochastic parameter c_n is given by the method proposed in Ref. ¹⁰:

$$f_{L_{n,j}}(l_{n,j} | c_n) = \frac{\omega_{n,j} - v(l_{n,j}) + c_n \lambda_0 l_{n,j} \mu(l_{n,j} + t_{n,j} - T_n, \theta)}{\sqrt{2\pi\sigma_B^2 l_{n,j}^3}} \times \exp \left\{ -\frac{(\omega_{n,j} - v(l_{n,j}))^2}{2\sigma_B^2 l_{n,j}} \right\} \quad (5)$$

where $v(l_{n,j}) = c_n \lambda_0 \int_{t_{n,j}-T_n}^{t_{n,j}+l_{n,j}-T_n} \mu(\tau, \theta) d\tau$, and $\omega_{n,j} = \omega - x_{n,j}$.

Considering the stochastic nature of the parameter c_n , the unconditional PDF of the RUL can be analytically derived based on the law of total probability. To facilitate the derivation process, the following lemma should be provided.

Lemma 1 ¹⁰. If $Z \sim N(\mu, \sigma^2)$, and $\omega_1, \omega_2, A, B \in \mathbb{R}$, $C \in \mathbb{R}^+$, then

$$E_Z \left[(\omega_1 - AZ) \exp \left\{ -\frac{(\omega_2 - BZ)^2}{2C} \right\} \right] = \sqrt{\frac{C}{B^2 \sigma^2 + C}} \left(\omega_1 - A \frac{B\omega_2 \sigma^2 + \mu C}{B^2 \sigma^2 + C} \right) \exp \left(-\frac{(\omega_2 - B\mu)^2}{2(B^2 \sigma^2 + C)} \right) \quad (6)$$

The proof of Lemma 1 can be deduced by generalizing the result in Ref. ¹⁰, and thus it is omitted here. Based on the above lemma, we can obtain the analytical PDF of the RUL at time $t_{n,j}$.

Proposition 1. For the stochastic degradation process defined in Eq. (1), based on the concept of the FHT, the analytical PDF of the RUL at time $t_{n,j}$ can be obtained by

$$f_{L_{n,j}}(l_{n,j}) = \sqrt{\frac{1}{2\pi l_{n,j}^2 (\lambda_0^2 h(l_{n,j})^2 \sigma_c^2 + \sigma_B^2 l_{n,j})}} \cdot (\omega_{n,j} - \lambda_0 \beta(l_{n,j}) u(l_{n,j})) \cdot \exp \left(-\frac{(\omega_{n,j} - \lambda_0 h(l_{n,j}) n u_c)^2}{2(\lambda_0^2 h(l_{n,j})^2 \sigma_c^2 + \sigma_B^2 l_{n,j})} \right) \quad (7)$$

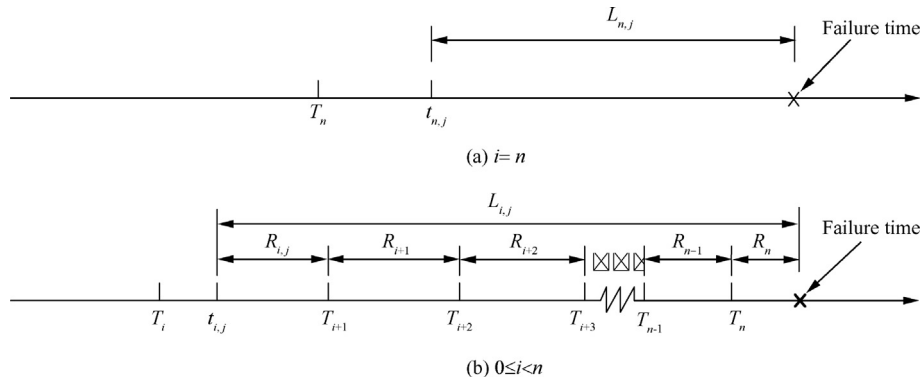


Fig. 2 Relationship between RUL and defined variables.

where $\beta(l_{n,j}) = \int_{l_{n,j}-T_n}^{l_{n,j}+l_{n,j}-T_n} \mu(\tau, \theta) d\tau - l_{n,j}\mu(l_{n,j} + l_{n,j} - T_n, \theta)$, $h(l_{n,j}) = \int_{l_{n,j}-T_n}^{l_{n,j}+l_{n,j}-T_n} \mu(\tau, \theta) d\tau$, and $u(l_{n,j}) = \frac{\sigma_c^2 \lambda_0 h(l_{n,j}) \omega_{n,j} + \sigma_B^2 l_{n,j} u_c n}{\lambda_0^2 h(l_{n,j})^2 \sigma_c^2 + \sigma_B^2 l_{n,j}}$.

Proof. Based on the law of total probability, the PDF of the RUL at time $t_{n,j}$ can be written as

$$f_{L_{n,j}}(l_{n,j}) = \int_{\Omega_n} f_{L_{n,j}}(l_{n,j}|c_n) f(c_n) dc_n \quad (8)$$

where Ω_n is the parameter space of c_n , $f(c_n)$ is the PDF of c_n , and $f_{L_{n,j}}(l_{n,j}|c_n)$ is the conditional PDF of the RUL at time $t_{n,j}$.

Substituting Eq. (5) into Eq. (8), we have

$$\begin{aligned} f_{L_{n,j}}(l_{n,j}) &= \int_{\Omega_n} f_{L_{n,j}}(l_{n,j}|c_n) f(c_n) dc_n \\ &= E_{c_n} \left(\frac{\omega_{n,j} - v(l_{n,j}) + c_n \lambda_0 l_{n,j} \mu(l_{n,j} + l_{n,j} - T_n, \theta)}{\sqrt{2\pi \sigma_B^2 l_{n,j}^3}} \right. \\ &\quad \times \exp \left\{ -\frac{(\omega_{n,j} - v(l_{n,j}))^2}{2\sigma_B^2 l_{n,j}} \right\} \Bigg) \end{aligned} \quad (9)$$

Let $\omega_1 = \frac{\omega_{n,j}}{\sqrt{2\pi \sigma_B^2 l_{n,j}^3}}$, $A = \frac{\lambda_0 (\int_{l_{n,j}-T_n}^{l_{n,j}+l_{n,j}-T_n} \mu(\tau, \theta) d\tau - l_{n,j}\mu(l_{n,j} + l_{n,j} - T_n, \theta))}{\sqrt{2\pi \sigma_B^2 l_{n,j}^3}}$, $\frac{\lambda_0 \beta(l_{n,j})}{\sqrt{2\pi \sigma_B^2 l_{n,j}^3}}$, $\omega_2 = \omega_{n,j}$, $B = \lambda_0 \int_{l_{n,j}-T_n}^{l_{n,j}+l_{n,j}-T_n} \mu(\tau, \theta) d\tau = \lambda_0 h(l_{n,j})$, $C = \sigma_B^2 l_{n,j}$, and so

Eq. (7) can be readily obtained based on Eqs. (6) and (9).

This completes the proof of Proposition 1.

Case 2. When $0 \leq i < n$, according to the PDF of the RUL defined in Eq. (4), to derive the PDF of the RUL, it is necessary to derive the PDF $f_{R_{i,j}}(r_{i,j})$ of the remaining operating time for the $(i+1)$ th stage, the PDF $f_{R_{i+k}}(r_{i+k})$ of the operating time for the $(i+k+1)$ th stage, and the PDF $f_{R_n}(r_n)$ of the operating time for the $(n+1)$ th stage. The proposition is given as follows to obtain the PDF $f_{R_{i,j}}(r_{i,j})$, and the conditional PDFs $f_{R_{i+k}}(r_{i+k}|\eta_{i+k})$, and $f_{R_n}(r_n|\eta_n)$.

Proposition 2. For the stochastic degradation process defined in Eq. (1), based on the concept of the FHT, we have the following results:

- (1) The PDF $f_{R_{i,j}}(r_{i,j})$ of the remaining operating time for the $(i+1)$ th stage can be denoted by

$$\begin{aligned} f_{R_{i,j}}(r_{i,j}) &= \sqrt{\frac{1}{2\pi r_{i,j}^2 (\lambda_0^2 h(r_{i,j})^2 \sigma_c^2 + \sigma_B^2 r_{i,j})}} (\omega_{i,j} - \lambda_0 \beta(r_{i,j}) u(r_{i,j})) \\ &\quad \exp \left(-\frac{(\omega_{i,j} - \lambda_0 h(r_{i,j}) u_c)^2}{2(\lambda_0^2 h(r_{i,j})^2 \sigma_c^2 + \sigma_B^2 r_{i,j})} \right) \end{aligned} \quad (10)$$

- (2) The conditional PDF $f_{R_{i+k}}(r_{i+k}|\eta_{i+k})$ of the operating time for the $(i+k+1)$ th stage can be denoted by

$$\begin{aligned} f_{R_{i+k}}(r_{i+k}|\eta_{i+k}) &= \sqrt{\frac{1}{2\pi r_{i+k}^2 (\lambda_0^2 h(r_{i+k})^2 \sigma_c^2 + \sigma_B^2 r_{i+k})}} \\ &\quad \times (\omega_{i+k} - \lambda_0 \beta(r_{i+k}) u(r_{i+k})) \\ &\quad \times \exp \left(-\frac{(\omega_{i+k} - \lambda_0 h(r_{i+k}) (i+k) u_c)^2}{2(\lambda_0^2 h(r_{i+k})^2 \sigma_c^2 + \sigma_B^2 r_{i+k})} \right) \end{aligned} \quad (11)$$

- (3) The conditional PDF $f_{R_n}(r_n|\eta_n)$ of the operating time for the $(n+1)$ th stage can be denoted by

$$\begin{aligned} f_{R_n}(r_n|\eta_n) &= \sqrt{\frac{1}{2\pi r_n^2 (\lambda_0^2 h(r_n)^2 \sigma_c^2 + \sigma_B^2 r_n)}} (\omega_n - \lambda_0 \beta(r_n) u(r_n)) \\ &\quad \times \exp \left(-\frac{(\omega_n - \lambda_0 h(r_n) n u_c)^2}{2(\lambda_0^2 h(r_n)^2 \sigma_c^2 + \sigma_B^2 r_n)} \right) \end{aligned} \quad (12)$$

where $1 \leq k < n-i$, $\omega_{i,j} = \omega_p - x_{i,j}$, $h(r_{i,j}) = \int_{t_{i,j}-T_i}^{t_{i,j}+r_{i,j}-T_i} \mu(\tau, \theta) d\tau$, $\beta(r_{i,j}) = \int_{t_{i,j}-T_i}^{t_{i,j}+r_{i,j}-T_i} \mu(\tau, \theta) d\tau - r_{i,j}\mu(t_{i,j} + r_{i,j} - T, \theta)$, $u(r_{i,j}) = \frac{\sigma_c^2 \lambda_0 h(r_{i,j}) \omega_{n,j} + \sigma_B^2 r_{i,j} u_c i}{\lambda_0^2 h(r_{i,j})^2 \sigma_c^2 + \sigma_B^2 r_{i,j}}$, $\omega_{i+k} = \omega_p - \eta_{i+k} \omega_p$, $h(r_{i+k}) = \int_0^{r_{i+k}} \mu(\tau, \theta) d\tau$, $\beta(r_{i+k}) = \int_0^{r_{i+k}} \mu(\tau, \theta) d\tau - r_{i+k}\mu(r_{i+k}, \theta)$, $u(r_{i+k}) = \frac{\sigma_c^2 \lambda_0 h(r_{i+k}) \omega_{i+k} + \sigma_B^2 r_{i+k} u_c (i+k)}{\lambda_0^2 h(r_{i+k})^2 \sigma_c^2 + \sigma_B^2 r_{i+k}}$, $\omega_n = \omega - \eta_n \omega_p$, $h(r_n) = \int_0^{r_n} \mu(\tau, \theta) d\tau$, $\beta(r_n) = \int_0^{r_n} \mu(\tau, \theta) d\tau - r_n \mu(r_n, \theta)$, $u(r_n) = \frac{\sigma_c^2 \lambda_0 h(r_n) \omega_n + \sigma_B^2 r_n u_c n}{\lambda_0^2 h(r_n)^2 \sigma_c^2 + \sigma_B^2 r_n}$, η_{i+k} denotes the residual degradation coefficient after the $(i+k)$ th maintenance activity, and η_n denotes the residual degradation coefficient after the n th maintenance activity.

The proof process is similar to the proof of Proposition 1, and hence we omit it here.

Due to the stochastic characteristics of η_{i+k} and η_n in the corresponding PDFs, Eqs. (11) and (12) denote the conditional PDFs, which have difference with Eq. (10). Hence, we can utilize the law of total probability to derive the PDFs $f_{R_{i+k}}(r_{i+k}) = E_{\eta_{i+k}} [f_{R_{i+k}}(r_{i+k}|\eta_{i+k})]$, and $f_{R_n}(r_n) = E_{\eta_n} [f_{R_n}(r_n|\eta_n)]$. Owing to the fact that the conditional PDFs $f_{R_{i+k}}(r_{i+k}|\eta_{i+k})$ and $f_{R_n}(r_n|\eta_n)$ are too complex to derive the theoretical expressions, the Monte-Carlo (MC) algorithm is adopted to calculate the PDFs $f_{R_{i+k}}(r_{i+k})$ and $f_{R_n}(r_n)$.

Based on Proposition 1, Proposition 2 and Eq. (4), we can obtain the PDF of the RUL for the equipment. The following lemma should be provided as the basis.

Lemma 2³¹. If $Y_1, Y_2, \dots, Y_n$ are the independent random variables with the corresponding PDFs $f_{Y_1}(y_1), f_{Y_2}(y_2), \dots, f_{Y_n}(y_n)$, then the PDF of the random variable $Z_n = \sum_{i=1}^n Y_i$ can be denoted by

$$\begin{aligned} f_Z(z) &= f_{Y_1}(y_1) \otimes f_{Y_2}(y_2) \otimes \dots \otimes f_{Y_n}(y_n) \\ &= \int \int \dots \int f_{Y_1}(y_1) f_{Y_2}(y_2) \dots f_{Y_{i-1}}(y_{i-1}) f_{Y_{i+1}}(y_{i+1}) \dots f_{Y_n}(y_n) \\ &\quad f_{Y_i}(z - y_1 - \dots - y_{i-1} - y_{i+1} - \dots - y_n) dy_1 \dots dy_{i-1} dy_{i+1} \dots dy_n \end{aligned} \quad (13)$$

where f_{Y_i} is one of the PDFs $f_{Y_1}(y_1), f_{Y_2}(y_2), \dots, f_{Y_n}(y_n)$.

The proof of Lemma 1 can be deduced by generalizing the result in Ref. ³¹, and thus it is omitted here. Furthermore, based on the Lemma 2, the PDF of the RUL of the equipment can be denoted by

$$f_{L_{ij}}(l_{ij}) = \int \int \cdots \int f_{R_{ij}}(r_{ij}) f_{R_{i+1}}(r_{i+1}) \cdots f_{R_n}(r_n) (l_{ij} - r_{ij} - \cdots - r_n) dr_{ij} dr_{i+1} \cdots dr_{n-1} \quad (14)$$

$0 \leq i < n$

Substituting the expressions of the PDFs $f_{R_{ij}}(r_{ij})$, $f_{R_{i+k}}(r_{i+k})$ ($1 \leq k < n - i$), and $f_{R_n}(r_n)$ into Eq. (14), we can obtain the PDF of the RUL of the equipment. The calculation of Eq. (14) involves multiple integral operations, which is so complicated that we cannot obtain the analytical expression. To obtain the PDF of RUL, a numerical simulation algorithm is formulated in this paper to calculate the PDF of the RUL. Specifically, MC simulation algorithm is also used to calculate the PDF $f(l_{ij})$ of the RUL at any time t_{ij} ($0 \leq i < n$) as follows.

Step 1. If $0 \leq i < n$, we can obtain the PDFs $f_{R_{ij}}(r_{ij})$, $f_{R_{i+k}}(r_{i+k}|\eta_{i+k})$ ($1 \leq k < n - i$), and $f_{R_n}(r_n|\eta_n)$ based on Eqs. (10)–(12) respectively.

Step 2. Select a sufficiently large positive integer (denoted by M_1), we can sample from the PDFs $f(\eta_{i+k})$ and $f(\eta_n)$ of residual degradation coefficient respectively. The p th specific sampling values are denoted by (η_{i+k}^p, η_n^p) , $0 \leq p \leq M_1$. The PDFs $f_{R_{i+k}}(r_{i+k})$ and $f_{R_n}(r_n)$ can be written as

$$f_{R_{i+k}}(r_{i+k}) \approx \frac{1}{M_1} \sum_{p=1}^{M_1} f_{R_{i+k}}(r_{i+k}|\eta_{i+k}^p) \quad (15)$$

$$f_{R_n}(r_n) \approx \frac{1}{M_1} \sum_{p=1}^{M_1} f_{R_n}(r_n|\eta_n^p) \quad (16)$$

Step 3. Select a sufficiently large positive integer (denoted by M_2) again and simulate the realizations from the PDFs $f_{R_{ij}}(r_{ij})$, $f_{R_{i+k}}(r_{i+k})$ ($1 \leq k < n - i$). We denote the p th specific realization by $(R_{ij}^p, R_{i+1}^p, \dots, R_{n-1}^p)$, where $0 \leq p \leq M_2$.

Step 4. For the sufficiently large number M_2 , based on the law of large numbers, we can approximately obtain the PDF $f_{L_{ij}}(l_{ij})$ of the RUL at any time t_{ij} :

$$f_{L_{ij}}(l_{ij}) \approx \frac{1}{M_2} \sum_{p=1}^{M_2} f_{R_n} \left(l_{ij} - R_{ij}^p - \sum_{k=1}^{n-i-1} R_{i+k}^p \right) \quad (17)$$

In order to have a better understanding of the MC simulation algorithm, a flowchart of the MC simulation algorithm is provided in Fig. 3.

4. Parameter estimation

The parameters that are needed to be estimated include the following three parts: residual degradation coefficient, degradation parameters and maintenance parameters.

4.1. Residual degradation coefficient estimation

The residual degradation coefficient can be estimated through the Maximum Likelihood Estimation (MLE) method.^{32,33} It is assumed that there are m groups of historical equipment which are independent for each other. Let $\mathbf{X}_{1:n}^s$ denote the residual degradation data of the s th ($1 \leq s \leq m$, $s \in \mathbb{N}^+$) historical equipment after each activity, where $\mathbf{X}_{1:n}^s = \{x_{1,0}^s, x_{2,0}^s, \dots, x_{n,0}^s\}$, and s is the number of the s th historical equipment. According to the degradation process defined in Eq. (1), the relation between the residual degradation coefficient and the residual degradation of the s th equipment at the i th PM activity can be given by

$$\eta_i^s = x_{i,0}^s / \omega_p \quad (18)$$

Based on Eq. (18) and the residual degradation of the s th equipment after each activity, we can obtain the residual degradation coefficient $\eta_{1:n}^s$ of the s th equipment after each PM activity, where $\eta_{1:n}^s = \{\eta_1^s, \eta_2^s, \dots, \eta_n^s\}$. From Eq. (1), we can find that the parameters in the distribution of the residual degradation coefficient η_i include a and b . The estimation of the parameters a and b is realized by the MLE. The log-likelihood function $\ell(a, b)$ is given by

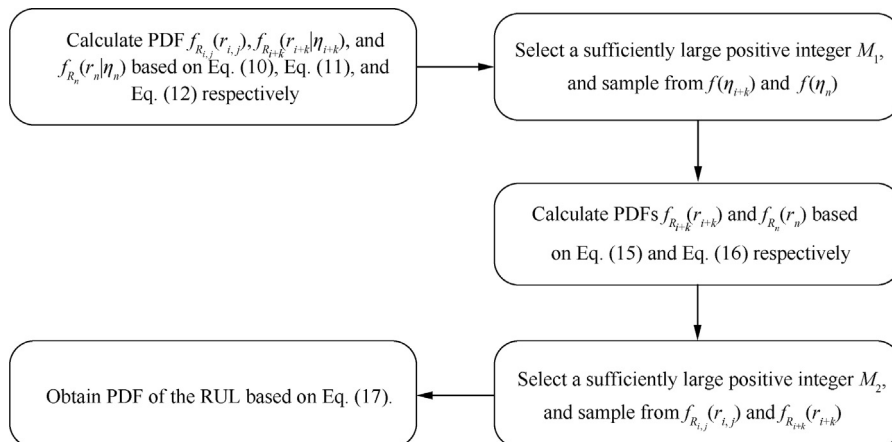


Fig. 3 Flowchart of proposed MC simulation algorithm.

$$\ln \ell(a, b) = -\frac{mn}{2} \ln 2\pi - \frac{mn}{2} \ln b - \frac{\sum_{s=1}^m \sum_{i=1}^n \left[(x_{i,0}^s / \omega_p) - (1 - \exp(-ai)) \right]^2}{2b} \quad (19)$$

In order to maximize the log-likelihood function in Eq. (19), we take the partial derivative of $\ln \ell(a, b)$ with respect to b :

$$\frac{\partial \ln \ell}{\partial b} = -\frac{mn}{2b} + \frac{\sum_{s=1}^m \sum_{i=1}^n \left[(x_{i,0}^s / \omega_p) - (1 - \exp(-ai)) \right]^2}{2b^2} \quad (20)$$

Let the partial derivative equal zero, and we have

$$b = \frac{1}{mn} \sum_{s=1}^m \sum_{i=1}^n \left[(x_{i,0}^s / \omega_p) - (1 - \exp(-ai)) \right]^2 \quad (21)$$

The log-likelihood function $\ln \ell(a)$ can be deduced by taking Eq. (21) into the log-likelihood function Eq. (19). By maximizing $\ln \ell(a)$, we can obtain the MLE of parameter a . Further, the estimation of b can be obtained by substituting the estimation of a into Eq. (21).

4.2. Degradation parameters estimation

The degradation parameters are also estimated utilizing m groups of historical degradation data. We suppose that all the observed degradation data of the s th historical equipment is denoted by $\mathbf{X}_{0:n_m}^s = \{x_{0,0}^s, x_{0,1}^s, x_{0,2}^s, \dots, x_{n,r_{n-1}}^s, x_{n,r_n}^s\}$. Based on the above assumptions, the degradation changing factor c_0 for the first stage equals one, while the degradation changing factor c_i for the $(i+1)$ th stage follows a normal distribution, i.e., $c_i \sim N(i\mu_c, \sigma_c^2)$. In order to make full use of m groups of historical degradation data, we should analyze each stage respectively.

For the first stage, based on Eq. (1) and the basic properties of the diffusion process, we have

$$\{x_{0,j}^s | x_{0,j-1}^s, \lambda_0, \theta, \sigma_B\} \sim N \left(x_{0,j-1}^s + \lambda_0 \int_{t_{0,j-1}}^{t_{0,j}} \mu(\tau, \theta) d\tau, \sigma_B^2 (t_{0,j} - t_{0,j-1}) \right)$$

and hence, the conditional PDF of $\{x_{0,j}^s | x_{0,j-1}^s, \lambda_0, \theta, \sigma_B\}$ can be written as

$$p(x_{0,j}^s | x_{0,j-1}^s, \lambda_0, \theta, \sigma_B) = \frac{1}{\sqrt{2\pi\sigma_B^2(t_{0,j} - t_{0,j-1})}} \cdot \exp \left(-\frac{(x_{0,j}^s - x_{0,j-1}^s - \lambda_0 \int_{t_{0,j-1}}^{t_{0,j}} \mu(\tau, \theta) d\tau)^2}{2\sigma_B^2(t_{0,j} - t_{0,j-1})} \right) \quad (22)$$

According to the Bayesian chain principle and the Markov property of the diffusion process, we have

$$\begin{aligned} \ell_0(\lambda_0, \theta, \sigma_B) &= p(\mathbf{X}_{0:r_0}^s | \lambda_0, \theta, \sigma_B) \\ &= p(x_{0,1}^s | \lambda_0, \theta, \sigma_B) p(x_{0,2}^s | x_{0,1}^s, \lambda_0, \theta, \sigma_B) \\ &\quad \cdots p(x_{0,r_0}^s | x_{0,r_0-1}^s, \lambda_0, \theta, \sigma_B) \\ &= \prod_{s=1}^m \prod_{j=1}^{r_0} \frac{1}{\sqrt{2\pi\sigma_B^2(t_{0,j} - t_{0,j-1})}} \\ &\quad \cdot \exp \left(-\frac{(x_{0,j}^s - x_{0,j-1}^s - \lambda_0 \int_{t_{0,j-1}}^{t_{0,j}} \mu(\tau, \theta) d\tau)^2}{2\sigma_B^2(t_{0,j} - t_{0,j-1})} \right) \end{aligned} \quad (23)$$

where $\ell_0(\lambda_0, \theta, \sigma_B)$ denotes the likelihood function for the first stage.

Similarly, for the $(i+1)$ th stage, the conditional PDF of $\{x_{i,j}^s | x_{i,j-1}^s, \lambda_0, \theta, \sigma_B, c_i\}$ can be written as

$$p(x_{i,j}^s | x_{i,j-1}^s, \lambda_0, \theta, \sigma_B, c_i) = \frac{1}{\sqrt{2\pi\sigma_B^2(t_{i,j} - t_{i,j-1})}} \cdot \exp \left(-\frac{(x_{i,j}^s - x_{i,j-1}^s - c_i \lambda_0 \int_{t_{i,j-1}}^{t_{i,j}} \mu(\tau, \theta) d\tau)^2}{2\sigma_B^2(t_{i,j} - t_{i,j-1})} \right) \quad (24)$$

Owing to the unknown characteristics of c_i , it is difficult to employ MLE to estimate the degradation parameters. According to Lemma 1, the conditional PDF of $\{x_{i,j}^s | x_{i,j-1}^s, \lambda_0, \theta, \sigma_B, \mu_c, \sigma_c\}$ can be given by

$$p(x_{i,j}^s | x_{i,j-1}^s, \lambda_0, \theta, \sigma_B, \mu_c, \sigma_c) = E_{c_i} [p(x_{i,j}^s | x_{i,j-1}^s, \lambda_0, \theta, \sigma_B, c_i)] = \frac{1}{\sqrt{2\pi Q}} \cdot \exp \left(-\frac{(x_{i,j}^s - x_{i,j-1}^s - i\mu_c \lambda_0 \int_{t_{i,j-1}}^{t_{i,j}} \mu(\tau, \theta) d\tau)^2}{2Q} \right) \quad (25)$$

where $Q = (\lambda_0 \int_{t_{i,j-1}}^{t_{i,j}} \mu(\tau, \theta) d\tau)^2 \sigma_c^2 + \sigma_B^2 (t_{i,j} - t_{i,j-1})$. Hence, the likelihood function $\ell_i(\lambda_0, \theta, \sigma_B, \mu_c, \sigma_c)$ for the $(i+1)$ th stage can be denoted by

$$\ell_i(\lambda_0, \theta, \sigma_B, \mu_c, \sigma_c) = \prod_{s=1}^m \prod_{j=0}^{r_i} \frac{1}{\sqrt{2\pi Q}} \cdot \exp \left(-\frac{(x_{i,j}^s - x_{i,j-1}^s - i\mu_c \lambda_0 \int_{t_{i,j-1}}^{t_{i,j}} \mu(\tau, \theta) d\tau)^2}{2Q} \right) \quad (26)$$

Owing to the assumption that every stage of the degradation process is independent with each other, the likelihood function $\ell(\lambda_0, \theta, \sigma_B, \mu_c, \sigma_c)$ based on m groups of historical degradation data can be written as

$$\ell(\lambda_0, \theta, \sigma_B, \mu_c, \sigma_c) = \ell_0(\lambda_0, \theta, \sigma_B) \cdot \prod_{i=1}^n \ell_i(\lambda_0, \theta, \sigma_B, \mu_c, \sigma_c) \quad (27)$$

The MLE values of $(\lambda_0, \theta, \sigma_B^2, \mu_c, \sigma_c)$ can be obtained by maximizing Eq. (27). It is worth noting that the MLE values of (μ_c, σ_c) can be used as the prior distribution of maintenance parameters.

4.3. Maintenance parameters estimation

We suppose that the observed degradation data of the operating equipment between the time of the k th PM and the time t_{k,r_k} are denoted as $\mathbf{X}_{0:r_k}^k = \{x_{k,0}, x_{k,1}, \dots, x_{k,r_k}\}$, where k is the PM number of the operating equipment before the time t_{k,r_k} , and r_k is the CM number of the operating equipment after the k th PM. The following proposition is given to estimate the maintenance parameters (μ_c, σ_c^2) .

Proposition 3. The posterior estimates of the maintenance parameters (μ_c, σ_c^2) at t_{k,r_k} can be written as

$$\mu_{c,kr_k} = \frac{1}{k} \frac{\sum_{j=1}^{r_k} \frac{\lambda_0 \rho (x_{k,j} - x_{k,j-1})}{\sigma_B^2(t_{k,j} - t_{k,j-1})} + \frac{k\mu_{c,k0}}{\sigma_{c,k0}^2}}{\sum_{j=1}^{r_k} \frac{\lambda_0^2 \rho^2}{\sigma_B^2(t_{k,j} - t_{k,j-1})} + \frac{1}{\sigma_{c,k0}^2}} \quad (28)$$

$$\sigma_{c,kr_k}^2 = \frac{1}{\sum_{j=1}^{r_k} \frac{\lambda_0^2 \rho^2}{\sigma_B^2(t_{k,j} - t_{k,j-1})} + \frac{1}{\sigma_{c,k0}^2}} \quad (29)$$

where $\rho = \int_{t_{k,j-1}-T_k}^{t_{k,j}-T_k} \mu(\tau, \theta) d\tau$, $(\mu_{c,kr_k}, \sigma_{c,kr_k}^2)$ denotes the updated results of the distribution parameter of c_k at t_{k,r_k} , $(\mu_{c,k0}, \sigma_{c,k0}^2)$ denotes the prior distribution parameters of c_k , and these prior values can be obtained by the updated results of the distribution parameter c_{k-1} before the k th PM.

Proof. Based on the degradation model described in Eq. (1) and the basic characteristic of the diffusion process, we have

$$\{x_{k,j}|x_{k,j-1}, c_k\} \sim N\left(x_{k,j-1} + c_k \lambda_0 \int_{t_{k,j-1}-T_k}^{t_{k,j}-T_k} \mu(\tau, \theta) d\tau, \sigma_B^2(t_{k,j} - t_{k,j-1})\right)$$

By employing the Bayesian chain principle, we can obtain the joint probability density of $\mathbf{X}_{0:r_k}^k$ conditional on the stochastic parameter c_k :

$$p(\mathbf{X}_{0:r_k}^k | c_k) = \prod_{j=1}^{r_k} \frac{1}{\sqrt{2\pi\sigma_B^2(t_{k,j} - t_{k,j-1})}} \cdot \exp\left(-\frac{(x_{k,j} - x_{k,j-1} - c_k \lambda_0 \rho)^2}{2\sigma_B^2(t_{k,j} - t_{k,j-1})}\right) \quad (30)$$

Let the prior distribution of c_k be denoted by $p(c_k)$, whose mean and variance are $k\mu_{c,k0}$ and $\sigma_{c,k0}^2$ respectively. $(\mu_{c,k0}, \sigma_{c,k0}^2)$ can be obtained based on the degradation data before the k th PM.

According to Bayesian theorem, the posterior distribution of the stochastic parameter c_k can be given by

$$p(c_k | \mathbf{X}_{0:r_k}^k) \propto p(\mathbf{X}_{0:r_k}^k | c_k) p(c_k) \quad (31)$$

Taking $p(\mathbf{X}_{0:r_k}^k | c_k)$ and $p(c_k)$ into Eq. (31), we have

$$\begin{aligned} p(c_k | \mathbf{X}_{0:r_k}^k) &\propto \prod_{j=1}^{r_k} \frac{1}{\sqrt{2\pi\sigma_B^2(t_{k,j} - t_{k,j-1})}} \exp\left(-\frac{(x_{k,j} - x_{k,j-1} - c_k \lambda_0 \rho)^2}{2\sigma_B^2(t_{k,j} - t_{k,j-1})}\right) \\ &\quad \frac{1}{\sqrt{2\pi\sigma_{c,k0}^2}} \exp\left(-\frac{(c_k - k\mu_{c,k0})^2}{2\sigma_{c,k0}^2}\right) \\ &\propto \exp\left(-\left(\sum_{j=1}^{r_k} \frac{\lambda_0^2 \rho^2}{2\sigma_B^2(t_{k,j} - t_{k,j-1})} + \frac{1}{2\sigma_{c,k0}^2}\right) c_k^2 + \right. \\ &\quad \left. \left(\sum_{j=1}^{r_k} \frac{(x_{k,j} - x_{k,j-1}) \lambda_0 \rho}{\sigma_B^2(t_{k,j} - t_{k,j-1})} + \frac{k\mu_{c,k0}}{\sigma_{c,k0}^2}\right) c_k - \right. \\ &\quad \left. \sum_{j=1}^{r_k} \frac{(x_{k,j} - x_{k,j-1})^2}{2\sigma_B^2(t_{k,j} - t_{k,j-1})} + \frac{k^2 \mu_{c,k0}^2}{2\sigma_{c,k0}^2}\right) \\ &\propto \exp\left(-\frac{(c_k - k\mu_{c,kr_k})^2}{2\sigma_{c,kr_k}^2}\right) \end{aligned} \quad (32)$$

The updated results of (μ_c, σ_c^2) can be derived by Eq. (32), as shown in Eqs. (28) and (29).

This completes the proof. $\square$

The main procedure of the aforementioned parameter estimation process can be summarized as follows.

- (1) The parameters (a, b) are estimated by the MLE based on the observed residual degradation data of the historical equipment after each PM;
- (2) The parameters $(\lambda_0, \theta, \sigma_B^2)$ are estimated by the MLE based on the observed degradation data of the historical equipment;
- (3) The posterior parameters of (μ_c, σ_c^2) are updated based on the real-time obtained degradation data $\mathbf{X}_{0:r_k}^k$ by Eqs. (28) and (29), where the priors of (μ_c, σ_c^2) can be obtained based on the observed degradation data of the historical equipment.

In order to have an intuitive understanding of the parameter estimation and RUL estimation method, we provide a flowchart of the RUL estimation method proposed in the paper, as shown in Fig. 4.

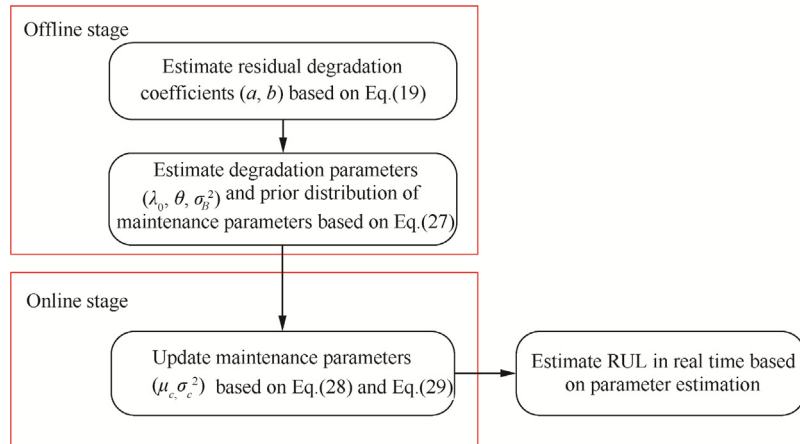


Fig. 4 Flowchart of proposed RUL estimation method.

Table 1 Parameter setting.

a	b	λ	θ	σ_B	μ_c	σ_c	n	ω_p	ω
0.2	0.001	1	1.5	0.2	1.6	0.03	4	20	25

5. Numerical example

This section verifies the effectiveness of the RUL estimating method proposed in the paper by a numerical example.

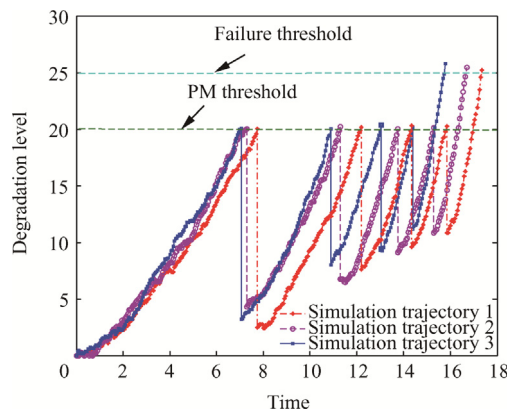
5.1. Data generation via simulation

In order to generate the simulated data, it is necessary to specify the parameters in Eq. (1). We set the parameters in Table 1 and select $\mu(\tau, \theta) = \theta\tau^{\theta-1}$ for simplification as used in the Ref. ¹⁰. Furthermore, we set the sampling time interval $\Delta t = 0.05$.

After setting the parameters, we generate three simulation degradation trajectories according to the degradation model defined in Eq. (1). The simulation degradation trajectories along with the PM threshold and the failure threshold are illustrated in Fig. 5. The simulation trajectories 1 and 2 as the historical equipment are used to estimate the residual degradation coefficient and the degradation parameters of the model defined in Eq. (1). The simulation trajectory 3 is used to update the maintenance parameters and estimate the RUL of the degradation process in real time.

5.2. Results and comparison

We can obtain the residual degradation after each maintenance for the simulation trajectories 1 and 2, i.e., $X_{1:4}^1 = \{x_{1,0}^1, x_{2,0}^1, x_{3,0}^1, x_{4,0}^1\}$ and $X_{1:4}^2 = \{x_{1,0}^2, x_{2,0}^2, x_{3,0}^2, x_{4,0}^2\}$.

**Fig. 5** Three simulation degradation trajectories.

Therefore, the residual degradation coefficient of the simulation trajectories 1 and 2 after each maintenance activity can be derived based on Eq. (18), i.e., $\eta_{1:4}^1$ and $\eta_{1:4}^2$. Table 2 provides the calculated residual degradation coefficient after each PM for the simulation trajectories 1 and 2.

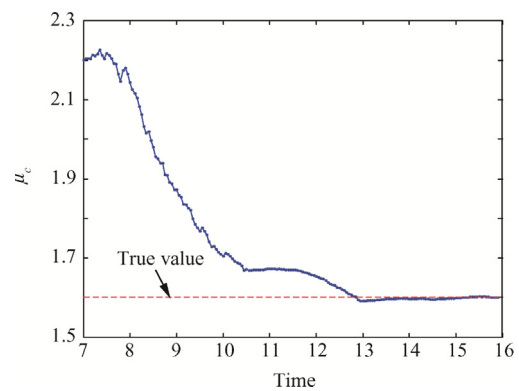
Therefore, the MLE value of a can be determined by taking the residual degradation coefficient into the log-likelihood function $\ln L(a)$. The estimated values of a and b are $\hat{a} = 0.2016$, and $\hat{b} = 0.0013$.

According to Section 4.2, we can obtain the MLE values of the degradation parameters ($\lambda_0, \theta, \sigma_B^2$) based on the simulation trajectories 1 and 2, as shown in Table 3. From Table 3, we can find that the estimation values for the degradation parameters have little difference with the true values, which verifies the effectiveness of the proposed degradation parameter estimation method.

Based on Proposition 3, the distribution parameters (μ_c, σ_c^2) can be updated online utilizing the degradation data of the simulation trajectory 3. The initial value ($\mu_{c,10}, \sigma_{c,10}^2$) can be selected according to the degradation data of the historical equipment. Figs. 6 and 7 illustrate the updated results. From

Table 3 Degradation parameter estimation.

Value	λ	θ	σ_B
Estimated	1.1308	1.6405	0.2120
True	1	1.5	0.2

**Fig. 6** Updated results of μ_c .**Table 2** Residual degradation coefficient for simulation trajectories 1 and 2.

η_1^1	η_2^1	η_3^1	η_4^1	η_1^2	η_2^2	η_3^2	η_4^2
0.2135	0.3122	0.4744	0.575	0.2135	0.3216	0.4517	0.5028

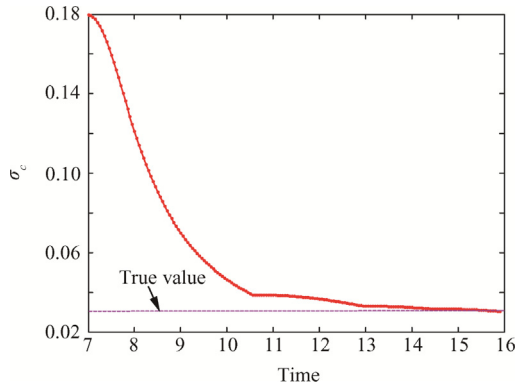


Fig. 7 Updated results of σ_c .

Figs. 6 and 7, we can find that the deviations between the initial values of (μ_c, σ_c^2) and their true values are large at the early stage because the historical sample is too few to characterize the truth of the parameters. However, with the acquisition of the degradation data, the estimation values of μ_c and σ_c are gradually approaching the true values, which verifies the effectiveness of the proposed maintenance parameter updating method.

After obtaining the estimated parameters, we select $t_{ij} = 7$ and $t_{ij} = 11$ as the time for RUL estimation. From the simulation trajectory 3, we can find that the first PM occurs at 6.95 and the second PM occurs at 10.95. Therefore, when $t_{ij} = 7$, there exists one PM activity before t_{ij} , i.e., $i = 1$, and there exists one CM datum after the first PM, i.e., $j = 1$. When $t_{ij} = 11$, there exist two PM activities before t_{ij} , i.e., $i = 2$, and there exists one CM datum after the second maintenance, i.e., $j = 1$. Based on Proposition 2, the PDF $f_{R_{ij}}(r_{ij})$ of the remaining operating time for the $(i + 1)$ th stage, the PDF $f_{R_{i+k}}(r_{i+k})$ of the operating time for the $(i + k + 1)$ th stage, and the PDF $f_{R_n}(r_n)$ of the operating time for the $(n + 1)$ th stage can be derived. After obtaining the PDFs $f_{R_{ij}}(r_{ij})$ and $f_{R_{i+k}}(r_{i+k})$ ($1 \leq k < n - i$), we can sample from the PDFs by the MC sampler. It is worth mentioning that the calculation time of the MC simulation algorithm at $t_{ij} = 7$ and $t_{ij} = 11$ is 45.96 s and 38.35 s respectively. In order to verify the effectiveness of the RUL estimation method proposed in the paper, the method that considers the influence of imperfect maintenance activities on both the degradation level and the degradation rate (i.e., the method proposed in this paper) is denoted by M_1 , the method ignoring the influence of imperfect maintenance activities on the degradation rate is denoted by M_2 (i.e., the method proposed in Ref. 22), while the method ignoring the influence of imperfect maintenance activities on the degradation level is denoted by M_3 (i.e., the method proposed in Ref. 23). Figs. 8 and 9 provide the PDFs of the RULs for the equipment at time 7 and time 11 for these three methods.

According to the above analysis, when $t_{ij} = 7$, we can sample from the PDFs $f(r_{1,1})$, $f(r_2)$, and $f(r_3)$. After we obtain the sampling results, the PDF of the RUL can be approximately calculated by Eq. (17), which is illustrated in Fig. 8. From Fig. 8, we can find that the estimated RUL by method M_1 is more accurate than that by the other two methods M_2 and M_3 . The estimated mean RULs for methods M_1 , M_2 , and

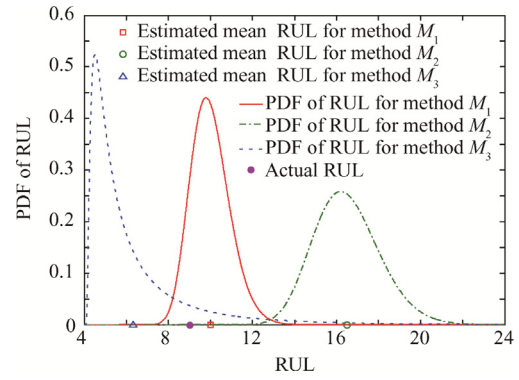


Fig. 8 PDF of RUL at $t_{ij} = 7$ for three methods.

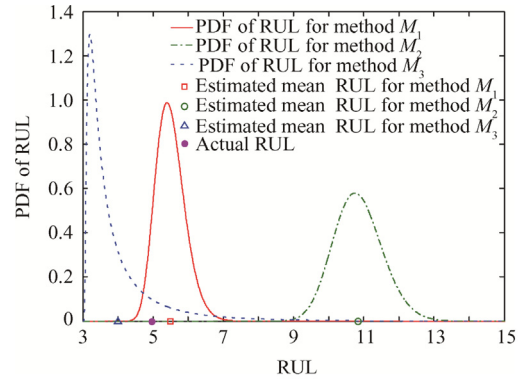


Fig. 9 PDF of RUL at $t_{ij} = 11$ for three methods.

M_3 are 10.0, 16.8, and 6.2 respectively, while the actual RUL is 8.9. So, method M_2 overestimates the RUL of the equipment, and the method M_3 underestimates the RUL of the equipment. Because methods M_2 and M_3 ignore the double influence of imperfect maintenance on the degradation level and the degradation rate. We can also find that the deviation between the estimated mean RUL for method M_1 and the actual RUL seems to be a bit large. This is because the degradation data obtained at $t_{ij} = 7$ is too limited, which leads to the uncertainty of the maintenance parameter estimation.

Similarly, when $t_{ij} = 11$, we can sample from the PDFs $f(r_{2,1})$ and $f(r_3)$. Based on the sampling results, the PDF of RUL can be approximately calculated by Eq. (17), as illustrated in Fig. 9. From Fig. 9, the estimated mean RULs for methods M_1 , M_2 , and M_3 are respectively 5.4, 10.8, and 4.0, while the actual RUL is 4.9. Obviously, the estimated mean RUL by method M_1 is more accurate than the estimated mean RULs by methods M_2 and M_3 . Owing to the fact that the CM data obtained at $t_{ij} = 11$ are increasing, the deviation between the estimated mean RUL by method M_1 and the actual RUL is gradually decreasing. However, the PDF of the RUL is calculated approximately by Eq. (17), and hence, there still exists a certain deviation between the estimated mean RUL of method M_1 and the actual RUL.

We have considered the situation of $0 \leq i < n$ above. In the following part, we will further consider the situation of $i = n$. Specifically, we select a series of time instants for RUL estima-

tion, i.e., {14.5, 14.6, 14.7, 14.8, 14.9, 15.0, 15.1, 15.2, 15.3, 15.4, 15.5}. Owing to the fact that the influence of imperfect maintenance activities on the degradation level can be reflected by the degradation data, the method that ignores the influence of imperfect maintenance activities on the degradation level (i.e., method M_3) will be obviously inappropriate for the RUL estimation here, and hence, this method is not employed for comparison. Fig. 10 provides the estimated PDFs and means of the RULs at the chosen estimation times for methods M_1 and M_2 .

From Fig. 10, we can find that the estimated mean RUL by method M_1 is close to the actual RUL, while the estimated mean RUL for method M_2 seriously deviates from the actual RUL. The main reason is that method M_2 ignores the influence of the imperfect maintenance activities on the system's degradation rate. Therefore, when method M_2 is used, it will overestimate the RUL of the equipment, which may lead to catastrophic accidents for the engineering equipment in practice.

To further compare the RUL estimation accuracy between these two RUL estimation methods quantitatively, the Mean

Squared Error (MSE) for the RUL estimation is adopted, which is defined as¹¹

$$\text{MSER}_j = \int_0^\infty (l_j - \tilde{l}_j)^2 f(l_j) dl_j \quad (33)$$

where $\tilde{l}_j$ is the actual RUL at t_{nj} , and $f(l_j)$ is the PDF of the RUL at t_{nj} . The reason that we choose the criterion defined in Eq. (30) to evaluate the RUL estimation accuracy is that this criterion can take the RUL estimation uncertainty into account, which is more reasonable than the other commonly-used criteria.¹¹

Based on Eq. (33), we can obtain the MSE at each estimation time for these two methods, which is shown in Fig. 11. We can see clearly from Fig. 11 that the MSE of the RUL estimation by method M_1 is far less than the MSE of the RUL estimation by method M_2 . As a result, the double influence of imperfect maintenance activities on degradation level and degradation rate should be taken into consideration while modeling the degradation and estimating the RUL of the equipment subjected to the intervention of imperfect maintenance activities.

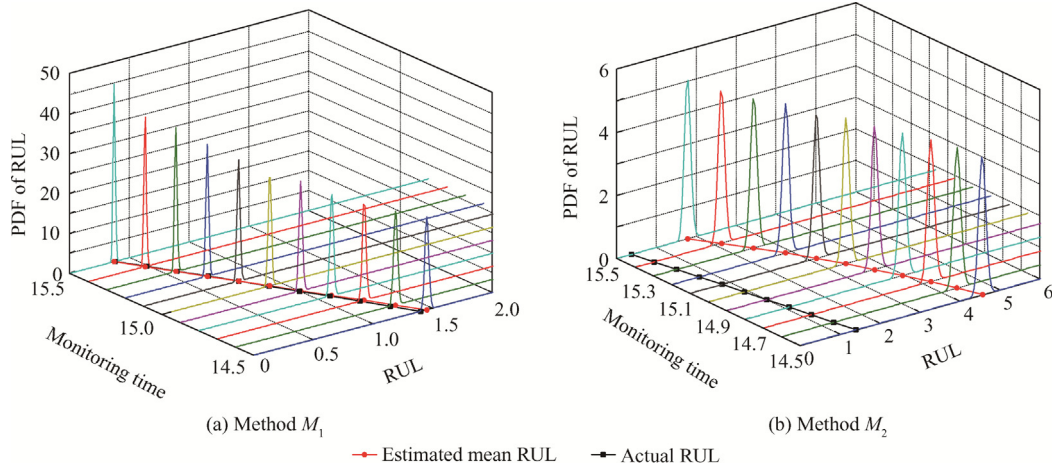


Fig. 10 Estimated PDFs and means of RULs of the gyro by method M_1 and M_2 .

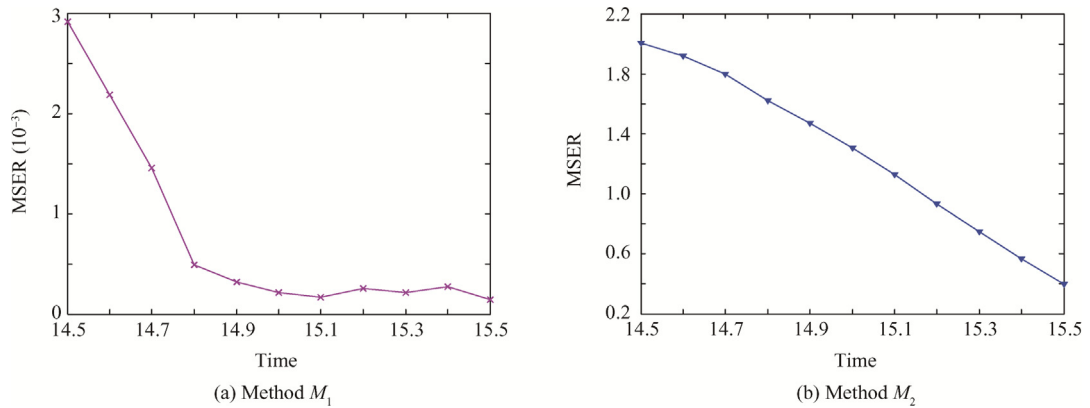


Fig. 11 MSE of estimated RULs at chosen times.

6. Practical case study

This section provides a practical case study of gyroscopes (abbreviated as gyros in this paper) in the inertial navigation systems to demonstrate the effectiveness and superiority of the proposed degradation modeling and RUL estimating method for deteriorating system subjected to the intervention of imperfect maintenance activities.

6.1. Problem description

As a key component of the inertial navigation systems, the gyro directly affects the precision of the navigated systems, such as missile, aircraft. In engineering, the performance of gyros inevitably degrades owing to the wear of the rotor spin axis and the creep deformation of the floated gimbal assembly. Generally, the degradation level of the gyro is characterized by its drift calibration values, called drift for short in this paper. So the drift of the gyro can be used as a performance indicator to evaluate the health condition of the gyro. When the drift of the gyro exceeds the preset PM threshold, the torque coil of the gyro will be adjusted to revise the drift of the gyro, which lowers the drift of the gyro and prolongs the useful life of the gyro. The adjustment can be regarded as the imperfect maintenance activity for the gyro. When the drift of the gyro exceeds the preset failure threshold, it means that the gyro becomes invalid. Hence, it must be replaced.

Fig. 12 illustrates two groups of gyroscopic drift data subjected to the intervention of imperfect maintenance activities. The drift data of the gyro 1 are utilized to estimate the parameters in the degradation modeling, while the drift data of the gyro 2 are utilized to update the maintenance parameter and estimate the RUL of the gyro. For our monitored inertial navigation systems, the gyro is tested periodically for 2.5 h each time. The PM threshold ω_p is 0.30 ($^{\circ}$)/h (i.e., degree per hour), and the failure threshold is 0.37 ($^{\circ}$)/h. As shown intuitively in Fig. 12, the actual life of the gyro1 is 277.5 h, and the actual

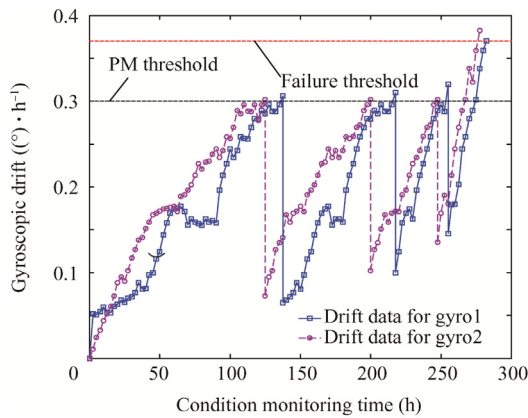


Fig. 12 Degradation paths of gyros.

life of the gyro 2 is 282.5 h. On the one hand, the drift cannot return to zero after each PM, i.e., there exists the residual degradation after each PM. On the other hand, the degradation speed increases after each PM to some extent. These two aspects of change verify that the PM can be regarded as imperfect maintenance with influence on both the degradation level and the degradation rate. The upper limit of the maintenance number performed in the gyro's life cycle is set to be three here.

6.2. Results and comparison

Similarly, we can obtain the MLE values of the parameters $(a, b, \lambda_0, \theta, \sigma_B^2)$ based on the simulation trajectory 1, as shown in Table 4. The estimation process of maintenance parameter is also similar to the last section, and hence, the maintenance parameter estimation is omitted here.

In order to verify the effectiveness of the RUL estimation method proposed in the paper, we select $t_{ij} = 150$ h as the time for RUL estimation. According to the drift data for gyro 2, the first PM occurs at 125 h, and therefore, when $t_{ij} = 150$ h, there exists one PM activity before t_{ij} . Based on the MC simulation algorithm proposed in the paper, Fig. 13 provides the PDF of the RUL of the gyro 2 at $t_{ij} = 150$ h for three methods.

From Fig. 13, we can also find that the RUL of the gyro 2 is overestimated by method M_2 , and the RUL of gyro 2 is underestimated by the method M_3 . In other words, the estimated RUL by method M_1 is the best among these three methods, which illustrates the superiority of the method proposed in the paper.

We have considered the situation of $0 \leq i < n$ above. In the following part, we will further consider the situation of $i = n$. Specifically, we select the time sequences as the RUL estimation points from 260 h to 280 h. We can obtain the PDFs of the RULs for gyro 2 by utilizing methods M_1 and M_2 . Fig. 14 shows the estimated PDFs and means of the RULs of the gyro by methods M_1 and M_2 .

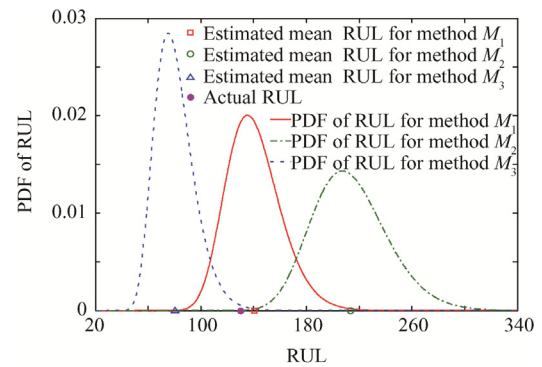


Fig. 13 PDF of RUL at $t_{ij} = 150$ h for three methods.

Table 4 Parameter estimation of gyros.

Parameter	a	b	λ_0	θ	σ_B
Estimated value	0.1916	0.0012	2.35×10^{-3}	0.946	1.61×10^{-4}

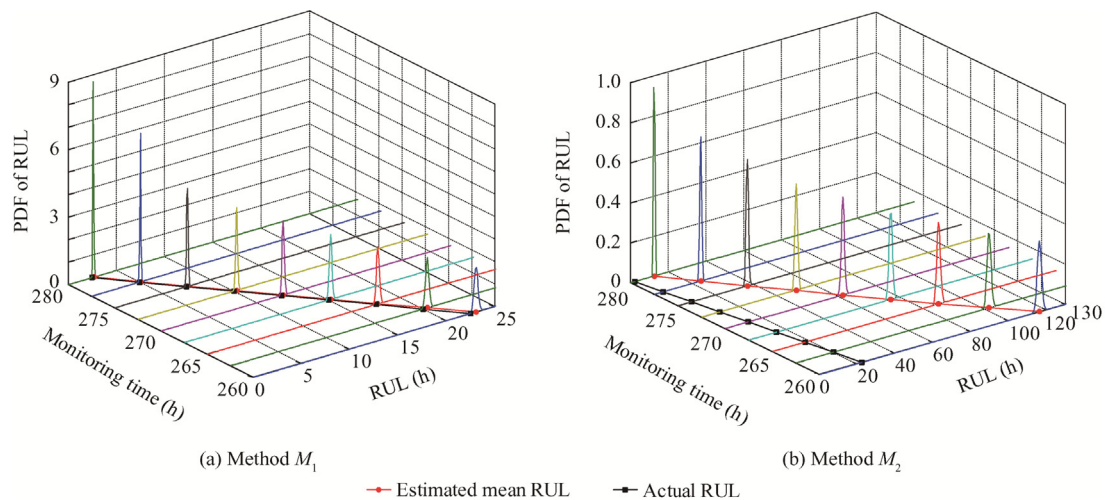


Fig. 14 Estimated PDFs and means of RULs.

It can be readily seen from Fig. 14 that the deviation between the estimated mean RUL by method M_2 and the actual RUL seems so large that it cannot accurately estimate the RUL of the gyro. However, the deviation between the estimated mean RUL by method M_1 and the actual RUL can be ignored after 265 h, and hence, the estimated mean RUL by method M_1 can be used to reflect the actual RUL of the gyro, which further demonstrates the effectiveness and superiority of the RUL estimation method proposed in the paper. From the above discussion, we can conclude that it is necessary to consider the double influence of imperfect maintenance on the degradation level and the degradation rate for the RUL estimation of the gyro subjected to the intervention of imperfect maintenance activities.

7. Conclusions

For the equipment subjected to imperfect maintenance activities, this paper presented a degradation modeling and RUL estimation method considering the influence of imperfect maintenance activities on both degradation level and degradation rate. Specifically, a stochastic degradation model considering the influence of imperfect maintenance activities was established based on the diffusion process model. The parameters in the degradation model were estimated by MLE and Bayesian updating strategy based on the CM data and the maintenance data of the equipment. Further, the PDF of RUL was derived by the convolution operator under the concept of FHT. A numerical example and a practical case study were used to verify the effectiveness and superiority of the method proposed in the paper. From the experimental results, we can find that the method proposed in the paper can estimate the RUL accurately for the equipment subjected to the intervention of imperfect maintenance activities, which is beneficial to the optimal maintenance and spare parts ordering. Different from the existing work, our method considers the influence of the imperfect maintenance on both the degradation level and the degradation rate, which is more consistent with the actual situations.

This is a preliminary study and there are several further research topics which will be studied in the future. The first

is to consider other distributions to characterize the degradation rate changing factor according to specific application background. The second is to make the maintenance or inventory decisions based on the prediction information obtained in this paper, since the main goal of the RUL estimation is to optimize the maintenance time and spare parts' ordering quantity by minimizing the operating cost or mitigating failure risk.

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